nature portfolio

Peer Review File

Genome sequencing of 'Fuji' apple clonal varieties reveals genetic mechanism of the spur-type morphology

Corresponding Author: Dr Zhangjun Fei

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors present an analysis of the Fuji apple cultivar, where somatic variation is analyzed by way of bud sport sequencing. Inferences about important traits are made, and a few are validated with follow up work. The Fuji reference genome is fully-phased and haplotype resolved. A phylogenetic analysis reveals potential bud sports that provide the basis for small clades with similar agronomically important traits (spur bearing, color, variability in horticultural maturity).

Over all the work seems sound methodologically, but issues are apparent in the presentation of the work.

line 109-10: there are many more "...previously published apple genomes..." than the ones they cite, certainly since the double haploid 'Golden Delicious' report in 2017 that they do include. The wording of the sentence implies that this is an exhaustive list, when recent fully phased Malus genomes from the last few years (perhaps the best points of comparison (HiFi + Hi-C/Omni-C) are not well-represented. The list they provide is incomplete, and the apparent size discrepancy of nearly 100Mbp between the 'Gala' haplomes from Sun et al is not mentioned. I recommend to add more citations and adjust the language accordingly. The Genome Database for Rosaceae is a good place to start.

Line 124: it is recommended to include the BUSCO score for the annotated genes as well. Recent work in Pyrus has shown that many genes were missing from a whole genome annotation, despite an expected number of annotated genes (Zhang et al 2022 - https://doi.org/10.3389/fpls.2022.975942). This was due to small assembly errors.

Line 208: Typo, should be sections?

There is a literature that addresses the topic of cell size/shape/density and internode length in apple, yet it is not cited. Though the authors do assert that their cytological data suggest a relationship between cell morphology and internode length at line 210. This is another example of an overlooked literature that has high relevance to the topic at hand - e.g. 10.1093/jxb/ern049.

What does it mean to "... instantly transform..." apple calli? Perhaps "immediately" should be used?

Line 293: Authors or researchers conclude, results do not. Perhaps "results indicate"

Line 322: The citation to the paper on willow is the incorrect citation to support a statement about Malus domestication history.

Line 322-3: High heterozygosity is actually a benefit to modern approaches of phased-genome assembly. The haplomes are easier to assemble.

Line 323: Further, other factors beyond the domestication history have contributed to high heterozygosity in apple.

Line 325: The "current apple reference genome" is a misnomer, especially now. There are many high quality apple genomes, and to exclude them here is improper. In fact there has been a Fuji genome out for months -

https://www.rosaceae.org/Analysis/15540493, which the authors cite subsequently.

These issues indicate that the work is not up to the level of Nature Communications in terms of rigor and careful contextualization, and polished presentation. A minimum threshold is to contextualize the results with the most recent and appropriate literature. I suggest to reject the paper, and urge the authors to resubmit after they have carefully revisited the relevant apple literature, and double checked their references.

Reviewer #2

(Remarks to the Author)
The authors constructed phased genomes of Fuji apple with long reads and short reads. With the assembled genomes, the authors then identified “somatic” mutations from 74 clonally propagated Fuji varieties, and characterized the potential functional somatic mutations, which leads to the discovery of a spur-type specific deletion within a promoter region of gene MdTCP11 that affects the expression of the gene, affecting the growth of the apple. While I found the findings overall are interesting, the assembled genomes are useful, and the sequenced data provides extra resources for the field, I have the following concerns and comments that the authors need to address:

1) The accuracy of the phased genomes is unknown.

Trio-based genome phasing approach was mainly developed and applied in animal genome assembly under assumption that: 1. the two haplotypes inherited from parents are almost the same as in the parents with only a small number of mutations accumulated; 2. the two haplotypes are different enough from each other. The authors applied the trio-based phasing approach on the Fuji genome assembly which is different: after many years of selection, the two haplotypes from the “child” genome have accumulated a large portion of mutations making them different from the parental haplotypes. Thus, how well the trio-based approach could be applied on the Fuji genome assembly is unclear. The evaluations the authors provide are mainly to evaluate the quality of the assembly, but no evaluation of the phasing quality. The authors need to provide extra evaluation of the phasing accuracy. For example, in the supplementary table, only hamming error rate is provided. The authors should also provide the switching error rate information. In Fig.S10, the authors showed haplotype specific kmers between their assembled Fuji_Ral and Fuji_Del haplotype. The authors also need to check: what's the distribution of these haplotype specific kmers in the parental sequencing data? Are these also ‘Delicious’ and ‘Ralls Janet’ specific? What’s the kmer-frequency distribution of these specific in ‘Delicious’ and ‘Ralls Janet’ sequencing data? The authors mentioned the phased assembly in citation 31, which is HiFi + HiC strategy based phasing approach. Although it is not separated for all the chromosomes, it will be helpful to compare the phasing accuracy between the two approaches on one or some chromosomes.

2) Mutation signature analysis performed in Fig.2 is interesting. It will be helpful to check whether there are different patterns between rare mutations (shared in less samples) and high allele frequency mutations to see whether there are different mutational mechanisms?
3) Germline variants are defined those heterozygous mutations shared by more than 80% of the Fuji varieties. Here, 80% is reasonable but still arbitrary. If the authors set different cutoff, say 50%, 60%, 70%, 80%, 90% and 100%, will they see a convergence?
4) Line 464-465, please provide detailed alignment score and edit distance cutoffs here when the authors exclude alignments.
5) For the genetic transformation, please include experiments to verify the transgene insertion and its overexpression.
6) If the transgenic apple plants have been transplanted to the greenhouse, please provide photos of them instead of in the bottle.
7) In Figure 6E, the OE-16 line exhibited a distinct dwarfing phenotype compared to the other two lines. Please provide an explanation for this difference.
8) Since the expression level of MdTCP11 was also higher in other spur-type varieties than in standard-type varieties, the 167-bp deletion may not be the sole reason for the altered expression of MdTCP11. Please refine your conclusion and speculate on other possible mechanisms.

Reviewer #3

(Remarks to the Author)
In this manuscript, Cai et al. have conducted a multidisciplinary study which identified a TCP transcription factor which play a crucial role in the developmental decision to produce standard-type or spur-type apple tree. After conducting a thorough genomic analysis comparing 74 somatic variant (clonally-propagated) they identified a spur-type and an early maturing-type clade. While they could not identify a somatic structural variant specific to the early maturing clade, they discovered that a deletion in the MITE region of the promoter of a TCP gene is found in the different varieties of the spur-type clade. Afterwards, they showed than the expression level of this transcription factor is lower in standard-type due the methylation of the promoter as a consequence of the presence of the transposon. They finally carried out a functional characterization confirming that MdTCP11 can indeed repress plant growth (Plant height and internode length) in Nicotiana and in apple (GL3).

This manuscript is well written, with results clearly presented and convincing. It nicely provides a new example confirming that TCP genes play an important role in controlling plant architecture.

Major comments:
- In my opinion, authors could have better taken advantage of their data. Some experiments are shown without any explanation of their scientific meaning and in consequence the biological question is not clear. Some data are also presented with no explanation and no discussion afterward, making them seem a bit useless. For instance, figure 4d shows some histological results but we do not know the purpose of this experiment, what tissue layer is observed, and the results are not discussed afterwards. How do the authors interpret these results? Is there a lack of cell differentiation in spur-type? Continuing with figure 4, the author showed that they found GA3 and GA4 (figure 4c), but they did not mention that cytokinin and ABA content were also significantly different between spur-type and standard-type which is also interesting. Both hormones can be regulated by TCP14/TCP15 (for review Nicolas and Cubas 2016) and have the potential to explain the observed phenotype. Cytokinin is well known to promote growth and ABA inhibits lateral root growth development in stress conditions (for review Du and Scheres 2018) and promote bud dormancy (González-Grandío et al., 2017). ABA could actually explain the cell size reduction observed in figure 4d in my opinion. I will be interested to see whether MdTCP11 expression is also affected by cytokinins and ABA like GA3 does (figure 6c). If they do, it would mean that a hormone cross-talk regulates MdTCP11 and the spur-type/standard-type regulation. The authors could also have introduced better why they consider GA to play a role upstream of MdTCP11 and not downstream (which was my first feeling).

- I am not a tree specialist and I was sometimes confused by the terminology used. I have a particular doubt. What do the author mean by shoot tip (transcriptomic/expression analyses and hormone measurement)? The term shoot is too vague for me. Is it the tip of the main shoot, the tip of the lateral branches or the tip of the spur-type or standard-type short shoot? Interpretation of the data is different depending of the type of tissue it is.

- It is a pity that these shoot tips have not been used to measure the methylation. Methylation patterns could vary with the tissues and it is not guaranteed that a similar methylation of MdTCP11 promoter will be observed in shoot tips. I would have been better to use the same kind of tissue for the expression and hormone and methylation analysis. Nevertheless, I can understand that working with trees have some technical limitations. Is it possible to easily determine the methylation of the spur/standard of the shoot?

- There are two classes of TCP, class I and class II. The canonical model states that class I TCP transcription factors promote cell growth and proliferation while class II proteins repress these processes (Li et al. 2005). The clade to which MdTCP11 belongs, including TCP11, TCP14, TCP15 is from the Class I. Moreover, TCP11, TCP14 and TCP15 consistently promote plant growth including plant height and internode length (Viola et al., 2011; Kiefer et al., 2011). It is thus surprising to see MdTCP11 inhibiting growth like a Class II TCP gene. This is possible and would an interesting result but that is why I would first like to have the guaranty that the methylation pattern and the expression pattern has been observed in the proper tissue. Moreover, I could not find in the methods how the TCP phylogenetic tree was built. Is it based on CDS or protein sequence alignment? In general, a phylogenetic tree based on the alignment of the TCP domain gives the best results when comparing sequences from phylogenetically distant species. In any case, it is important to mention the presence of the two TCP classes for the

- What is the length of the promoter used in the Dual Luciferase assays? I also found very surprising that 35S:Luc has the same level of expression as the TCp11∂MITE promoter in figure 6d. Do you have any explanation?

- What are the reference genes used for the qPCR?

- What is the difference between sup figure 8a and figure 5d? I believe it is “spur-type other” instead of clade in sup figure 8a. Data of supp figure 8b are not shown in the results and not discussed.

- What is the promoter used for the overexpression of MdTCP11 in nicotiana transgenic plant? Is it 35S? Results of the transgenic are convincing but are you sure this is Nicotiana benthamiana? Leaf shape looks very different from the roundish ones of Nicotiana benthamiana and this plants usually flower after 6 nodes. Could it be a different Nicotiana species? Maybe Nicotiana tabacum?

- Finally, the authors performed an RNA-seq but only used it to confirm the qPCR results of MdTCP11. I believe much more data could have been shown. How about showing an heatmap of GA, CK and ABA biosynthesis and signalling pathways gene expression level? Or expression of the Della proteins and possibly the different TCPs?

Minor comments:
- line 185: can you better explain what do you mean by early maturing?
- Figure 4: Check the quality of the figure. Median bars sometimes are missing, boxes are not rectangular, low image quality is sometimes quite low.
- Fig 4a: I imagine it is already Liquan spur and Yang No.8 used here
- Lines 208 and 533: “cell size” is more correct than “cell length”.
- There is a typo in sup table 13: standard.
- Line 298: typo apple.
- Please explain in the legend what are the blue and green boxes in Sup figure 7?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
My technical concerns have been addressed, and the authors have made some progress on my primary concern that apparently still persists from my initial review: "A minimum threshold is to contextualize the results with the most recent and appropriate literature."

The language at line 106-7 still implies that it is an exhaustive list by saying "...the recently published genome assemblies of domesticated apples." Simply remove the definite article and expand the list. As it is currently written, the passage is still inaccurate. Though the list is better, two important genomes are still missing, please add the 'Honeycrisp' genome: 10.46471/gigabyte.69 and the T2T 'Golden Delicious' genome: https://doi.org/10.1093/plphys/kiae258.

The language at line 325-330 is more concerning, stating directly that such apple genomes do not exist - in fact they do. See above.

The authors are still failing to acknowledge milestone apple genome resources, and the language at this passage implies an exhaustive list.

Reviewer #2

(Remarks to the Author)
The authors have addressed all my questions in the previous round and I don't have further comments. Congratulations to the authors!

Reviewer #3

(Remarks to the Author)
The authors have taken my recommendations into consideration and made the appropriate corrections. They also added results which I consider interesting to support the conclusions. In my opinion, the proposed new version is therefore suitable for publication.

Below are few minor mistakes or modifications that I believe should be corrected:

- Figure 5c, top of the Ga3 treated with 5-AZA is missing, at least the error bars are not present.

- Lines 281-287 and figure 6c: I missed this point in my last review. Conclusion of this experiments are not well presented. I think there are three mains important results in this experiment. One is that MdTCP11 expression responds to the 5-AZA treatment confirming that the methylation of the MITE promoter induces the reduction of MdTCP13 expression. The second is that GA3 reduces MdTCP11 which is further confirmed in figure 6d and sup fig 16. A lower concentration of GA3-4 in the spur tips (figure 4d) is thus consistent with a higher expression of MdTCP11. The third point is that 5-AZA activates MdTCP11 even in presence of GA3 suggesting the MITE region is not important for MdTCP11 regulation by GAs.

- Line 395, please cite a reference, for instance Martín-Trillo and Cubas 2010.

Reviewer #1:

The authors present an analysis of the Fuji apple cultivar, where somatic variation is analyzed by way of bud sport sequencing. Inferences about important traits are made, and a few are validated with follow up work. The Fuji reference genome is fully-phased and haplotype resolved. A phylogenetic analysis reveals potential bud sports that provide the basis for small clades with similar agronomically important traits (spur bearing, color, variability in horticultural maturity). Over all the work seems sound methodologically, but issues are apparent in the presentation of the work.

line 109-10: there are many more "...previously published apple genomes..." than the ones they cite, certainly since the double haploid 'Golden Delicious' report in 2017 that they do include. The wording of the sentence implies that this is an exhaustive list, when recent fully phased Malus genomes from the last few years (perhaps the best points of comparison (HiFi + Hi-C/Omni-C) are not well-represented. The list they provide is incomplete, and the apparent size discrepancy of nearly 100Mbp between the 'Gala' haplomes from Sun et al is not mentioned. I recommend to add more citations and adjust the language accordingly. The Genome Database for Rosaceae is a good place to start.
Response: Thank you for the suggestion. We have revised the manuscript by including recently published genomes of domesticated apples [1-7] (Line 106-107 and Supplementary Table 2).

References

1. Li, W. et al. Near-gapless and haplotype-resolved apple genomes provide insights into the genetic basis of rootstock-induced dwarfing. Nat Genet 56, 505-516 (2024)
2. Peng, H. et al. A haplotype-resolved genome assembly of *Malus domestica* ‘Red Fuji’. Sci Data 11, 592 (2024).
3. Su, Y. et al. Phased telomere-to-telomere reference genome and pangenome reveal an expansion of resistance genes during apple domestication. Plant Physiology 195, 2799-2814 (2024).
4. Wang, T. et al. Pan-genome analysis of 13 *Malus* accessions reveals structural and sequence variations associated with fruit traits. Nat Commun 14, 7377 (2023).
5. Daccord, N. et al. High-quality de novo assembly of the apple genome and methylome dynamics of early fruit development. Nat. Genet. 49, 1099 (2017).
6. Zhang, L. et al. A high-quality apple genome assembly reveals the association of a retrotransposon and red fruit colour. Nat Commun 10, 1494 (2019).
7. Sun, X. et al. Phased diploid genome assemblies and pan-genomes provide insights into the genetic history of apple domestication. Nat Genet 52, 1423–1432 (2020).

Line 124: it is recommended to include the BUSCO score for the annotated genes as well. Recent work in Pyrus has shown that many genes were missing from a whole genome annotation, despite an expected number of annotated genes (Zhang et al 2022 -

https://doi.org/10.3389/fpls.2022.975942). This was due to small assembly errors.
Response: Thanks. We have added the BUSCO scores for the annotated genes (99.2% for Fuji_Ral and 99.3% for Fuji_Del) in Supplementary Table 2 of the revised manuscript.

Line 208: Typo, should be sections?
Response: Corrected. Thanks.

There is a literature that addresses the topic of cell size/shape/density and internode length in apple, yet it is not cited. Though the authors do assert that their cytological data suggest a relationship between cell morphology and internode length at line 210. This is another example of an overlooked literature that has high relevance to the topic at hand - e.g. 10.1093/jxb/ern049.
Response: Thank you for pointing this out. We have cited the reference suggested by the reviewer in the revised manuscript (Line 213 and 219).

What does it mean to "... instantly transform..." apple calli? Perhaps "immediately" should be used?
Response: Corrected. Thanks.

Line 293: Authors or researchers conclude, results do not. Perhaps "results indicate"
Response: Corrected. Thanks.

Line 322: The citation to the paper on willow is the incorrect citation to support a statement about Malus domestication history.
Response: Thanks for pointing this out. In the revised manuscript, we have removed this sentence. Please refer to our responses to your comments below.

Line 322-3: High heterozygosity is actually a benefit to modern approaches of phased-genome assembly. The haplomes are easier to assemble.
Line 323: Further, other factors beyond the domestication history have contributed to high heterozygosity in apple.
Response: Thanks for the comments. We agree with the reviewer that other factors such as self-incompatibility, clonal propagation could have also contributed to the high heterozygosity in apple. Given the focus of this study, we have removed these two sentences in the revised manuscript.

Line 325: The "current apple reference genome" is a misnomer, especially now. There are many high quality apple genomes, and to exclude them here is improper. In fact there has been a Fuji genome out for months - https://www.rosaceae.org/Analysis/15540493, which the authors cite subsequently.
Response: Thank you for your suggestion. We have revised the manuscript to highlight our innovation in genome phasing and variant detection (Line 325-339).

These issues indicate that the work is not up to the level of Nature Communications in terms of rigor and careful contextualization, and polished presentation. A minimum threshold is to contextualize the results with the most recent and appropriate literature. I suggest to reject the paper, and urge the authors to resubmit after they have carefully revisited the relevant apple literature, and double checked their references.
Response: Thank you for the suggestion. We have carefully revised the manuscript in line with the reviewer's recommendations to enhance the presentation and contextualization.

Reviewer #2:

The authors constructed phased genomes of Fuji apple with long reads and short reads. With the assembled genomes, the authors then identified “somatic” mutations from 74 clonally propagated Fuji varieties, and characterized the potential functional somatic mutations, which leads to the discovery of a spur-type specific deletion within a promoter region of gene MdTCP11 that affects the expression of the gene, affecting the growth of the apple. While I found the findings overall are interesting, the assembled genomes are useful, and the sequenced data provides extra resources for the field, I have the following concerns and comments that the authors need to address:
1) The accuracy of the phased genomes is unknown.
Trio-based genome phasing approach was mainly developed and applied in animal genome assembly under assumption that: 1. the two haplotypes inherited from parents are almost the same as in the parents with only a small number of mutations accumulated; 2. the two haplotypes are different enough from each other. The authors applied the trio-based phasing approach on the Fuji genome assembly which is different: after many years of selection, the two haplotypes from the “child” genome have accumulated a large portion of mutations making them different from the parental haplotypes. Thus, how well the trio-based approach could be applied on the Fuji genome assembly is unclear. The evaluations the authors provide are mainly to evaluate the quality of the assembly, but no evaluation of the phasing quality. The authors need to provide extra evaluation of the phasing accuracy. For example, in the supplementary table, only hamming error rate is provided. The authors should also provide the switching error rate information. In Fig.S10, the authors showed haplotype specific kmers between their assembled Fuji_Ral and Fuji_Del haplotype. The authors also need to check: what's the distribution of these haplotype specific kmers in the parental sequencing data? Are these also ‘Delicious’ and ‘Ralls Janet’ specific? What's the kmer-frequency distribution of these specific in ‘Delicious’ and ‘Ralls Janet’ sequencing data? The authors mentioned the phased assembly in citation 31, which is HiFi + HiC strategy based phasing approach. Although it is not separated for all the chromosomes, it will be helpful to compare the phasing accuracy between the two approaches on one or some chromosomes.
Response: Thank you for your comments. We acknowledge the importance of evaluating the accuracy of the phased genome assemblies, especially given the differences between our study

and the typical trio-based genome phasing approach. First, we found that the average number of somatic mutations accumulated in each 'Fuji' individual is about 2,100 (2155 ± 935), which is substantially less than the variations observed between the two parent genomes (approximately 4.48 million). Additionally, kinship analysis confirmed that 'Fuji' samples can be accurately identified as the offspring of 'Delicious' and 'Ralls Janet' (Fig. 3b). Therefore, the two haplotypes of 'Fuji' have accumulated only a very small number of mutations after inherited from the parents. This is expected, given the short history of its breeding (created in 1930's; https://plants.ces.ncsu.edu/plants/malus-domestica-fuji/). Combined with the high heterozygosity of the 'Fuji' genome, which suggests significant difference between the two haplotypes, we conclude that it is appropriate to apply the trio-based method to the assembly of the 'Fuji' genome.

To further assess the quality of our phased genome assembly, we conducted a systematic k-mer-based quality evaluation, including a comparison with two recently released genome assemblies of 'Fuji' [1,2] (Supplementary Note and Supplementary Figs. 4-7). Analysis of k-mers from 'Delicious', 'Ralls Janet', and 'Fuji' raw sequencing data confirmed the high quality of our assembly, showing correct haplotype representation (Supplementary Fig. 7). We found that Fuji_Ral and Fuji_Del primarily contained haplotype-specific k-mers from 'Ralls Janet' and 'Delicious', respectively, unlike the other two assemblies, in which each haplome contained significant amounts of k-mers from both parents (Supplementary Fig. 5). Further haplotype-specific k-mer analysis revealed high phasing accuracy across all three genome assemblies, although in the other two assemblies each haplome showed mixing of chromosomes originating from the two parents (Supplementary Fig. 6). Furthermore, we found that the chromosome-level phasing accuracy was comparable between our assembly and the two other assemblies, with our assembly exhibiting 30 switch blocks affecting 1.96% of the genome (25.95Mb), compared to 39 and 34 switch blocks in the other two assemblies (29.62 Mb and 27.97Mb) (Supplementary Fig. 7). Overall, these results confirmed the high accuracy of our phased genome assembly, and the trio-based approach used for our assembly proved advantageous in correctly assigning chromosomes to their respective haplotypes. We have included these analysis results in the revised manuscript (Line 118-120 and Supplementary Note).

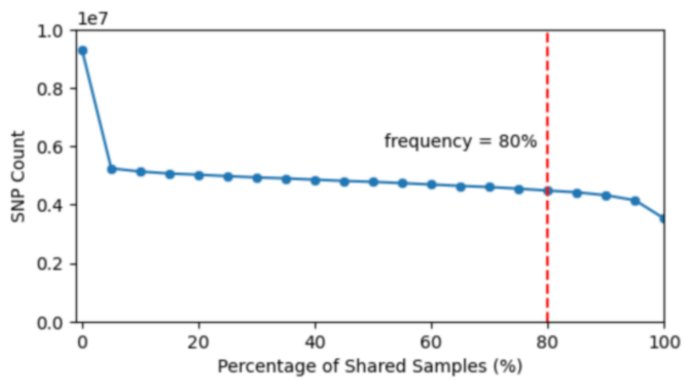
References

1. Li, W. et al. Near-gapless and haplotype-resolved apple genomes provide insights into the genetic basis of rootstock-induced dwarfing. Nat Genet 56, 505–516 (2024).
2. Peng, H. et al. A haplotype-resolved genome assembly of Malus domestica 'Red Fuji'. Sci Data 11, 592 (2024).

2) Mutation signature analysis performed in Fig.2 is interesting. It will be helpful to check whether there are different patterns between rare mutations (shared in less samples) and high allele frequency mutations to see whether there are different mutational mechanisms?

Response: Thank you for raising this interesting question. However, it is technically not feasible to perform the suggested analysis for high allele frequency germline variants because we cannot classify these mutations in the same way as low-frequency somatic variants. For example, in the case of a T/C mutation, we cannot determine whether it is a C->T or T->C mutation. We believe different mechanisms should be involved in the generation of low allele frequency somatic variants (which are spontaneous) and high allele frequency germline variants (which primarily reflect differences between parental genomes). However, the current data is insufficient to address this question, and we believe this is beyond the scope of the current study.

3) Germline variants are defined those heterozygous mutations shared by more than 80% of the Fuji varieties. Here, 80% is reasonable but still arbitrary. If the authors set different cutoff, say 50%, 60%, 70%, 80%, 90% and 100%, will they see a convergence?
Response: Thank you for your comment. To verify the threshold selection, we first calculated the number of germline variants identified using different thresholds (Response Fig. 4). We found that the number of identified variants remained consistent when applying thresholds between 20% and 80%. However, when the threshold exceeded 90%, a decline in the number of identified germline variants became apparent. Since the 80% threshold is near the end of this plateau, the number of identified germline variants is quite stable with respect to threshold selection. Thus, choosing thresholds within the plateau phase (including 80%) results in consistent outcomes, as shown in Response Fig. 1.



Response Figure 1 Number of identified germline SNPs (y-axis) using different thresholds (x-axis).

4) Line 464-465, please provide detailed alignment score and edit distance cutoffs here when the authors exclude alignments.
Response: Thank you for raising this question. We did not use a specific threshold for this analysis. Instead, we excluded alignments by comparing the alignment scores between genomes for the same pair of reads. For example, if a pair of reads has a higher alignment score and a lower edit distance against Fuji_Ral compared to Fuji_Del, we exclude this pair of reads from the Fuji_Del alignment file. We have clarified this in the revised manuscript (Line 475-478).

5) For the genetic transformation, please include experiments to verify the transgene insertion and its overexpression.
Response: Thank you for your valuable suggestion. We have included experiments to verify the transgene insertion and its overexpression in the revised manuscript (Supplementary Fig. 17**)**.

6) If the transgenic apple plants have been transplanted to the greenhouse, please provide photos of them instead of in the bottle.
Response: Thank you. We have added images of the *MdTCP11* overexpression lines and wild-type plants grown in the greenhouse for 90 days after transplanting (Fig. 6f of the revised manuscript).

7) In Figure 6E, the OE-16 line exhibited a distinct dwarfing phenotype compared to the other two lines. Please provide an explanation for this difference.
Response: Thank you for your comment. While the OE-16 plant shown in Fig. 6e exhibited a more severe dwarfing phenotype, the plant height and internode length of OE-16 lines (n = 5) were not statistically different from those of the other two lines (OE-17 and OE-20), as shown in the bar graphs of Fig. 6e. The observed difference in the OE-16 line could be attributed to its relatively higher expression level of *MdTCP11* compared to the other two lines, though this difference was not statistically significant, as shown in Supplementary Fig. 18b. This could also be due to differences among individual OE-16 plants. Unfortunately, while we measured the plant height and internode length for all OE-16 individuals, we did not photograph the other four OE-16 plants.

8) Since the expression level of MdTCP11 was also higher in other spur-type varieties than in standard-type varieties, the 167-bp deletion may not be the sole reason for the altered expression of MdTCP11. Please refine your conclusion and speculate on other possible mechanisms.
Response: Thank you for your comments. Since other spur-type varieties are dispersed across different clades in the phylogenetic tree, we speculate that different mutations may be responsible for the spur-type phenotype in these varieties. These causal variants might be located within the *MdTCP11* gene region, directly affecting its expression and function, or in upstream genes, such as transcription factors, that regulate *MdTCP11* expression. The precise molecular mechanisms underlying the spur-type phenotype in these varieties will require further investigation. We have discussed this in the revised manuscript (Line 413-420).

Reviewer #3:

In this manuscript, Cai et al. have conducted a multidisciplinary study which identified a TCP transcription factor which play a crucial role in the developmental decision to produce standard-type or spur-type apple tree. After conducting a thorough genomic analysis comparing 74 somatic variant (clonally-propagated) they identified a spur-type and an early maturing-type clade. While they could not identify a somatic structural variant specific to the early maturing clade, they discovered that a deletion in the MITE region of the promoter of a TCP gene is found in the different varieties of the spur-type clade. Afterwards, they showed than the expression level of this transcription factor is lower in standard-type due the methylation of the promoter as a consequence of the presence of the transposon. They finally carried out a functional characterization confirming that MdTCP11 can indeed repress plant growth (Plant height and internode length) in Nicotiana and in apple (GL3).

This manuscript is well written, with results clearly presented and convincing. It nicely provides a new example confirming that TCP genes play an important role in controlling plant architecture.

Major comments:

- In my opinion, authors could have better taken advantage of their data. Some experiments are shown without any explanation of their scientific meaning and in consequence the biological question is not clear. Some data are also presented with no explanation and no discussion afterward, making them seem a bit useless. For instance, figure 4d shows some histological results but we do not know the purpose of this experiment, what tissue layer is observed, and the results are not discussed afterwards. How do the authors interpret these results? Is there a lack of cell differentiation in spur-type? Continuing with figure 4, the author showed that they found GA3 and GA4 (figure 4c), but they did not mention that cytokinin and ABA content were also significantly different between spur-type and standard-type which is also interesting. Both hormones can be regulated by TCP14/TCP15 (for review Nicolas and Cubas 2016) and have the potential to explain the observed phenotype. Cytokinin is well known to promote growth and ABA inhibits lateral root growth development in stress conditions (for review Du and Scheres 2018) and promote bud dormancy (González-Grandío et al., 2017). ABA could actually explain the cell size reduction observed in figure 4d in my opinion. I will be interested to see whether MdTCP11 expression is also affected like GA3 does (figure 6c). If they do, it would mean that a hormone cross-talk regulates MdTCP11 and the spur-type/standard-type regulation. The authors could also have introduced better why they consider GA to play a role upstream of MdTCP11 and not downstream (which was my first feeling).

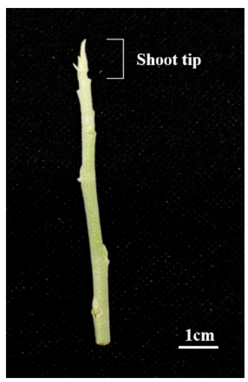
Response: Thank you for pointing this out. Cell differentiation plays a crucial role in internode elongation; therefore, we examined the morphological characteristics of cells in both spur-type and standard-type varieties, as shown in Fig. 4d. We have now explained these results in the revised manuscript (Line 213-219).

You are correct in suggesting that cytokinin and ABA may play a role in regulating the spur-type phenotype; therefore, we conducted an experiment to determine whether *MdTCP11* expression was affected by cytokinin and ABA. We analyzed the activities of the constitutive 35S promoter, MdTCP11-ΔMITE-pro, and MdTCP11-pro after cytokinin (6-BA) or ABA treatments. The results showed that LUC activities in both MdTCP11-ΔMITE-pro and MdTCP11-pro were significantly enhanced by ABA treatment, but not by 6-BA (Supplementary Fig. 16b). This suggests that a potential crosstalk between ABA and GA in regulating *MdTCP11* expression and the spur-type phenotype (Line 402-408).

Following your suggestions, we also measured GA_3 content in transgenic *N. tabacum* and apple lines overexpressing *MdTCP11*. The results indicated that GA_3 content was lower in both transgenic lines compared to wild-type plants, suggesting that gibberellin may be involved in a negative feedback regulatory mechanism governing the development of apple internodes. This experiment has been included in the revised manuscript (Line 397-401 and Supplementary Fig. 17).

- I am not a tree specialist and I was sometimes confused by the terminology used. I have a particular doubt. What do the author mean by shoot tip (transcriptomic/expression analyses and hormone measurement)? The term shoot is too vague for me. Is it the tip of the main shoot, the tip of the lateral branches or the tip of the spur-type or standard-type short shoot? Interpretation of the data is different depending of the type of tissue it is.

Response: Thanks. Shoot tips refer to apical portions (5-8 mm below the apical region, including the shoot meristem without any leaves or petioles; Response Fig. 2). We have clarified this in the revised manuscript (Line 533).



Response Figure 2. Shoot tip in an annual shoot.

- It is a pity that these shoot tips have not been used to measure the methylation. Methylation patterns could vary with the tissues and it is not guaranteed that a similar methylation of MdTCP11 promoter will be observed in shoot tips. I would have been better to use the same kind of tissue for the expression and hormone and methylation analysis. Nevertheless, I can understand that working with trees have some technical limitations. Is it possible to easily determine the methylation of the spur/standard of the shoot?

Response: Thank you for your valuable suggestion. We conducted a genome-wide analysis of cytosine methylation in shoot tips of ‘Liquan spur’ (a spur-type variety) and ‘Yanfu No.8’ (a standard-type variety) (Line 591-598). The results showed that the average DNA methylation level of the MITE in standard-type varieties was higher than that in spur-type varieties. This analysis has been included in the revised manuscript (Line 277-279 and Supplementary Fig. 15).

- There are two classes of TCP, class I and class II. The canonical model states that class I TCP transcription factors promote cell growth and proliferation while class II proteins repress these processes (Li et al. 2005). The clade to which MdTCP11 belongs, including TCP11, TCP14, TCP15 is from the Class I. Moreover, TCP11, TCP14 and TCP15 consistently promote plant growth including plant height and internode length (Viola et al., 2011; Kiefer et al., 2011). It is thus surprising to see MdTCP11 inhibiting growth like a Class II TCP gene. This is possible and would an interesting result but that is why I would first like to have the guaranty that the methylation pattern and the expression pattern has been observed in the proper tissue. Moreover, I could not find in the methods how the TCP phylogenetic tree was built. Is it based on CDS or protein sequence alignment? In general, a phylogenetic tree based on the alignment of the TCP domain gives the best results when comparing sequences from phylogenetically distant species. In any case, it is important to mention the presence of the two TCP classes for the

Response: Thank you. We used protein sequence to construct the phylogenetic tree, which indicates that MdTCP11 belongs to class I. In the revised manuscript, following the reviewer's recommendation, we constructed the phylogenetic tree based on the alignment of the TCP domain, which also supports that *MdTCP11* belongs to class I. We have now clarified the tree construction method in the legend of Fig. 5 (Line 248-249). We have also added a discussion on class I and class II TCP TFs in the revised manuscript (Line 392-397).

- What is the length of the promoter used in the Dual Luciferase assays? I also found very surprising that 35S:Luc has the same level of expression as the TCp11△MITE promoter in figure 6d. Do you have any explanation?

Response: Thank you for pointing this out. The 2000-bp MdTCP11 promoter sequence was used in the Dual Luciferase assays. We have clarified this in the revised manuscript (Line 558). While the results in our original manuscript did indicate that the activity of the MdTCP11△MITE promoter was lower than that of the 35S:LUC, this difference was not statistically significant. To address this, we repeated the experiment with more biological replicates (increased from 3 to 6) and found that the activity of the MdTCP11△MITE promoter was significantly lower compared to the 35S:LUC. We have updated the manuscript accordingly (Fig. 6d).

- What are the reference genes used for the qPCR?

Response: Thank you. The *ACTIN* gene was used as the reference (Supplementary Table 15). We have clarified this in the revised manuscript (Line 578).

- What is the difference between sup figure 8a and figure 5d? I believe it is "spur-type other" instead of clade in sup figure 8a. Data of supp figure 8b are not shown in the results and not discussed.

Response: We apologize for the unclear labeling. Fig. 5d compares *MdTCP11* expression levels between the spur-type 'Liquan spur' and the standard-type 'Yanfu No.8' across different developmental stages, while Fig. 8a shows qPCR results from various varieties (10 spur-type and 10 standard-type) at a single developmental stage. We have now clarified these details in the figure legends.

- What is the promoter used for the overexpression of MdTCP11 in nicotiana transgenic plant? Is it 35S? Results of the transgenic are convincing but are you sure this is Nicotiana benthamiana? Leaf shape looks very different from the roundish ones of Nicotiana benthamiana and this plants usually flower after 6 nodes. Could it be a different Nicotiana species? Maybe Nicotiana tabacum?

Response: Thank you for pointing this out. The 35S promoter was indeed used to overexpress *MdTCP11*. The *Nicotiana* variety used in transformation was *N. tabacum* L. cv. NC89. We have corrected this error throughout the revised manuscript.

- Finally, the authors performed an RNA-seq but only used it to confirm the qPCR results of MdTCP11. I believe much more data could have been shown. How about showing an heatmap of GA, CK and ABA biosynthesis and signalling pathways gene expression level? Or expression of the Della proteins and possibly the different TCPs?

Response: Thank you for your suggestion. Following the reviewer's recommendation, we analyzed the expression patterns of genes related to GA, CK, and ABA hormones using the RNA-Seq data. The results revealed that five genes related to GA, five genes related to CK, and one gene related to ABA were differentially expressed between the standard-type and spur-type varieties. These findings have been added to the revised manuscript (Line 403-405 and Supplementary Fig. 21).

Minor comments:

- line 185: can you better explain what do you mean by early maturing?

Response: In this study, "early-maturing varieties" refer to bud sport varieties whose fruit reaches the mature stage approximately 20 days earlier than the standard 'Fuji' variety. We have added this explanation to the Supplementary Table 10.

- Figure 4: Check the quality of the figure. Median bars sometimes are missing, boxes are not rectangular, low image quality is sometimes quite low.

Response: Fixed. Thanks.

- Fig 4a: I imagine it is already Liquan spur and Yang No.8 used here

Response: We apologize for the lack of clarity. The phenotypic data presented in Fig 4a are from various standard-type and spur-type 'Fuji' varieties. The specific 'Fuji' varieties used for measuring spur rate, crown width, and trunk height, as shown in Fig. 4a, are listed in Supplementary Table 14. We have clarified this in the legend of Fig. 4 in the revised manuscript (Line 204).

- Lines 208 and 533: "cell size" is more correct than "cell length".

Response: In this study, we only measured cell length, not cell size.

- There is a typo in sup table 13: standard.

- Line 298: typo apple.
Response: Corrected. Thanks.

- Please explain in the legend what are the blue and green boxes in Sup figure 7?
Response: Done. Thanks.

Reviewer #1:

My technical concerns have been addressed, and the authors have made some progress on my primary concern that apparently still persists from my initial review: “A minimum threshold is to contextualize the results with the most recent and appropriate literature.”

The language at line 106-7 still implies that it is an exhaustive list by saying “...the recently published genome assemblies of domesticated apples.” Simply remove the definite article and expand the list. As it is currently written, the passage is still inaccurate. Though the list is better, two important genomes are still missing, please add the ‘Honeycrisp’ genome: 10.46471/gigabyte.69 and the T2T ‘Golden Delicious’ genome: https://doi.org/10.1093/plphys/kiae258.

The language at line 325-330 is more concerning, stating directly that such apple genomes do not exist - in fact they do. See above.

The authors are still failing to acknowledge milestone apple genome resources, and the language at this passage implies an exhaustive list.
Response: Thank you for the suggestion. We have included these genomes in Supplementary Table 2, with citations on lines 100 and 277 in the revised version. Additionally, we updated the discussion of these literature on lines 275-277.

Reviewer #2:

The authors have addressed all my questions in the previous round and I don’t have further comments. Congratulations to the authors!
Response: Thanks.

Reviewer #3:

The authors have taken my recommendations into consideration and made the appropriate corrections. They also added results which I consider interesting to support the conclusions. In my opinion, the proposed new version is therefore suitable for publication.

Below are few minor mistakes or modifications that I believe should be corrected:

- Figure 5c, top of the Ga3 treated with 5-AZA is missing, at least the error bars are not present.
Response: Fixed. Thanks.

- Lines 281-287 and figure 6c: I missed this point in my last review. Conclusion of this experiments are not well presented. I think there are three mains important results in this experiment. One is that MdTCP11 expression responds to the 5-AZA treatment confirming that the methylation of the MITE promoter induces the reduction of MdTCP13 expression. The second is that GA3 reduces MdTCP11 which is further confirmed in figure 6d and sup fig 16. A lower concentration of GA3-4 in the spur tips (figure 4d) is thus consistent with a higher expression of MdTCP11. The third point is that 5-AZA activates MdTCP11 even in presence of GA3 suggesting the MITE region is not important for MdTCP11 regulation by GAs.

Response: Thank you for the suggestion. We have revised the text to better present the results (Line 240-241 and 252-257).

- Line 395, please cite a reference, for instance Martín-Trillo and Cubas 2010.

Response: Done. Thanks.